QPC Mini Project

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Michaelmas Term 2024

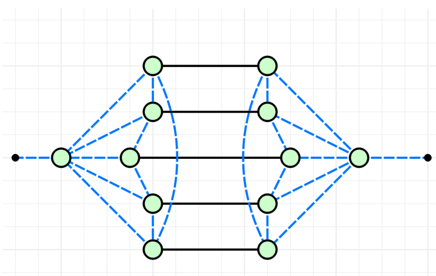
Problem 1.1

To show that E is an isometry, we need to verify the following identity:

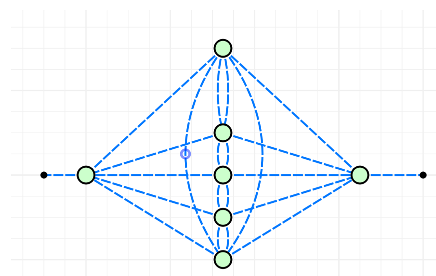
$$E^\dagger E = I.$$

By taking the conjugate of the diagram, we simplify the structure step by step until it reduces to the identity wire. The process involves the following transformations:

We begin by fusing the middle Z-spiders using spider fusion. The Z-spiders at the top and bottom each connect to a pair of Hadamard edges, which cancel each other out.

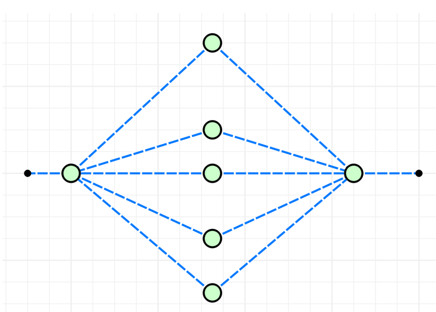


Step 1



Step 2

Since the Z spiders in the middle only have 1 input / output, they are equivalent to the identity wire, so they fall right off. The hadamard cancels out, leaving normal edge connections. This leaves two Z-spiders connected by 5 edges. Since even pairs of connections cancel out, only one connections remains. Consequently, the Z-spiders, now with just one input / output each, also falls off.



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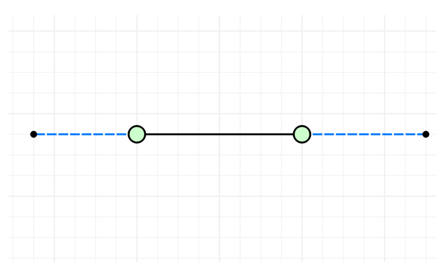
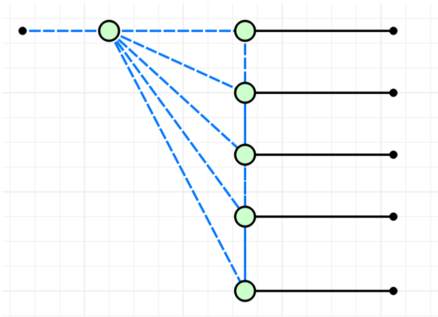


Step 4

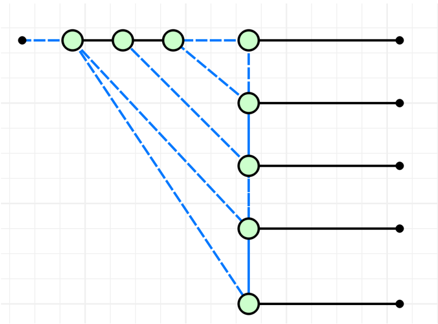
This leaves two Z-spiders connected by five edges. Since even pairs of connections cancel out, only one connection remains. Consequently, the Z-spiders, now with just one input/output each, also fall off. The hadamard gates on each side cancel out. This results in the identity wire: $E^\dagger E = I$. Therefore, E is an isometry.

Problem 1.2

Rewriting diagram E as a circuit consisting of only 1 or 2 qubit gates. We unfuse the Z-spiders to reduce the number of connections to a singular node.

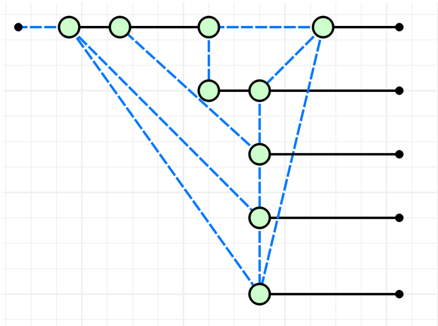


Step 1



Step 2

Some steps are skipped to keep the project compact, but the ideas is to keep unfusing the Z Spider on the top left with the most connections, as well as creating the ladder on the right.



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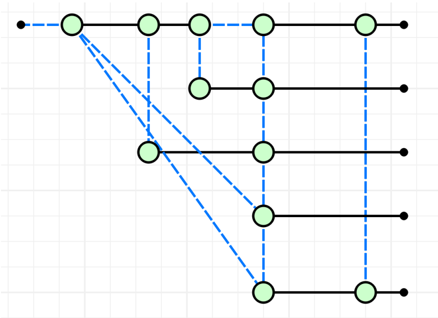

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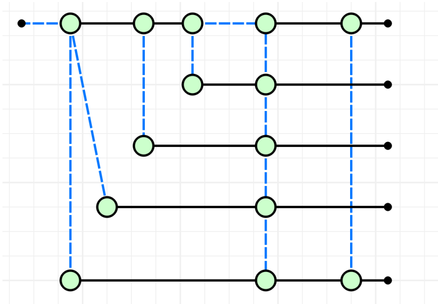

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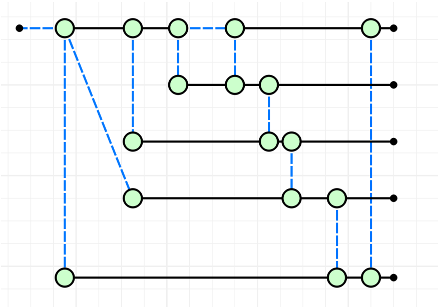
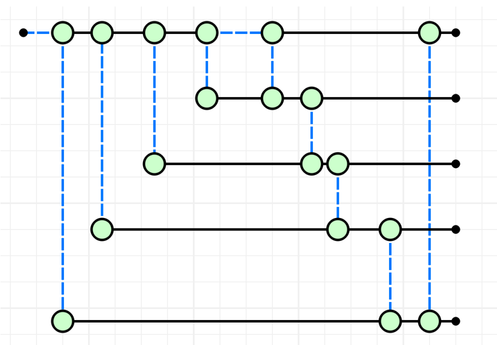
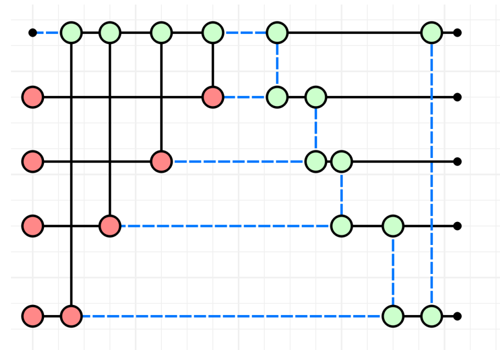


Step 6

Lastly, the auxilla qubits are added in the $|0\rangle$ state (which is an X spider with no phase), along with color change being applied to properly visualize the CNOT Gates. A ladder of CZ gates are on the right.



Step 7

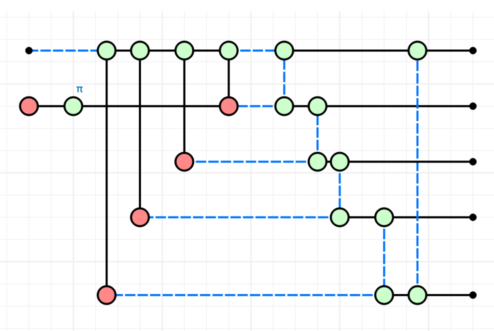


Step 8

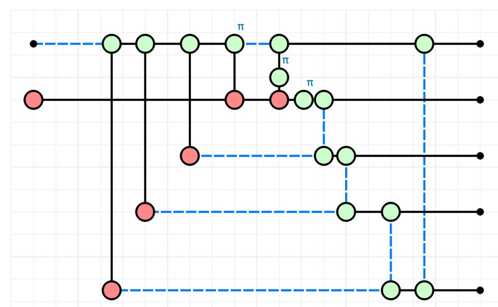
Problem 1.3

In this problem, we need to find the stabilizer generators. We have 1 logical qubit and 5 physical qubits, which means that there are $5 - 1 = 4$ stabilizer generators. There are 2 logical operators X, and Z. Since the auxiliary qubits are prepared at the $|0\rangle$ state, Pauli Z gates are introduced. This does not affect the logical qubit as adding the Pauli Z is diagrammatically equivalent, meeting the criterion of a stabilizer.

First Pauli String First, the Pauli Z gate is introduced on the top qubit. We then push it through to the end, to see how it'll affect the physical ones. Not all steps are enumerated, though the idea is the pi phase is pushed through the X-spider, then propogated until the nodes with phases are at the very end.



Step 1



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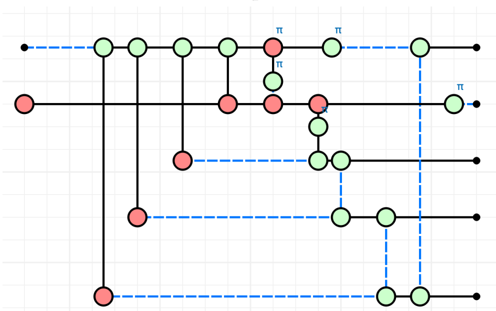

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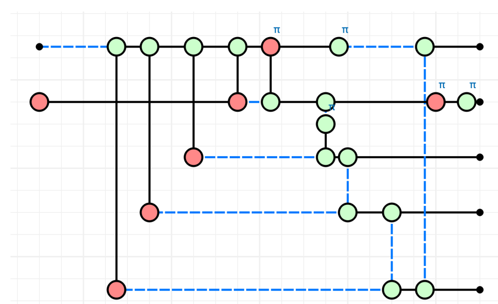

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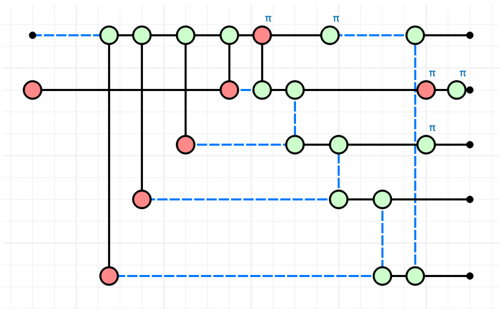
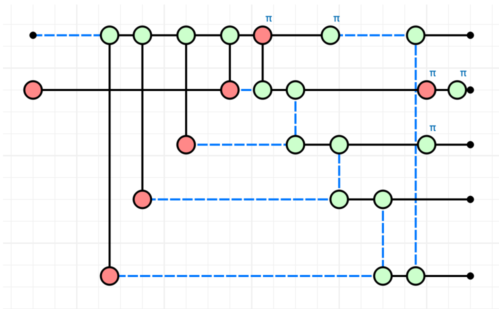


Step 5



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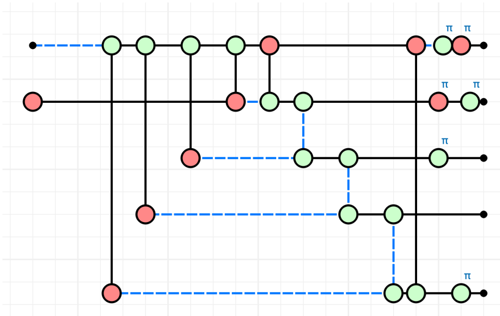
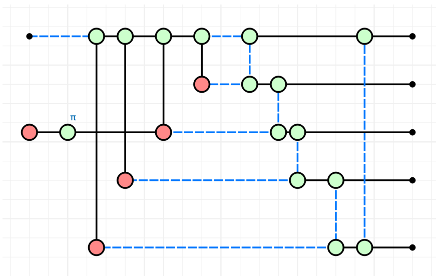


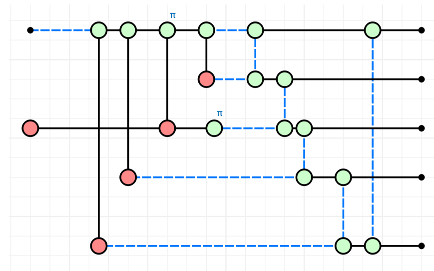
Step 7

After applying color change so the hadamards at the end are gone, we can see that a iY and -iY Pauli gates is on qubits 1 and 2. Qubits 3 and 5 have a Pauli Z gate, and the 4th qubit is the identity. Together, the first Pauli string is YYZIZ.

Second Pauli String The second stabilizer is found in the same way, by attaching a Pauli Z to the second qubit and propagating it through the diagram. Not all steps are shown to condense the figures.



Step 1



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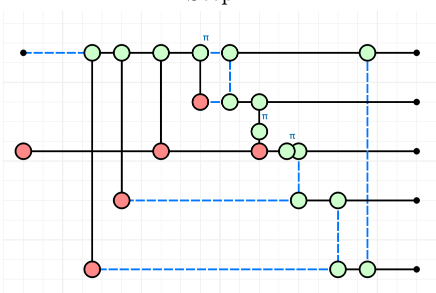

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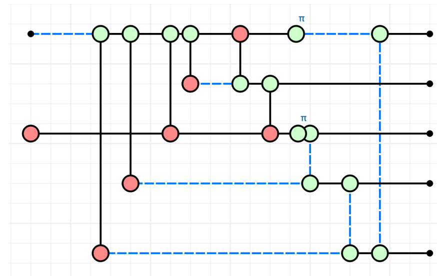

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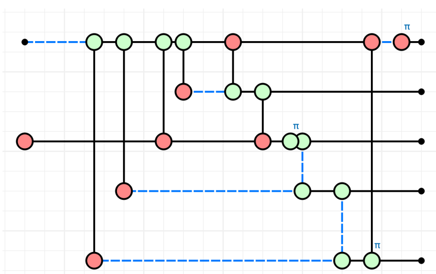

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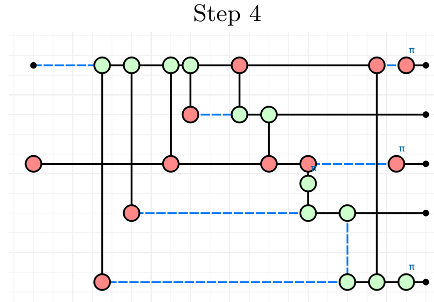

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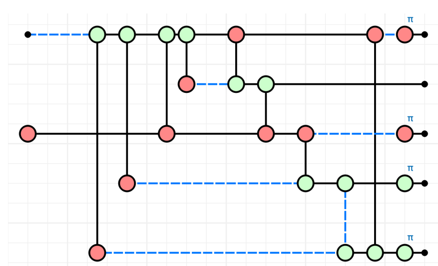
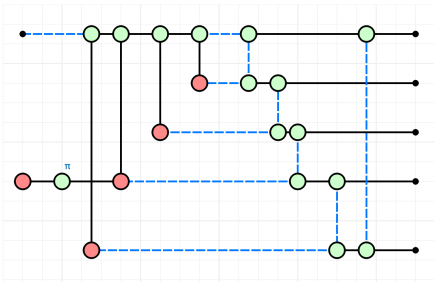


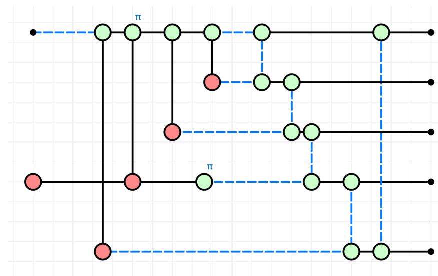
Step 7

In the end, there is an Pauli X on qubits 1 and 3, while qubits 4 and 5 have a Pauli Z gate. The 2nd qubit is the identity, which means the second Pauli String for the stabilizer is XIXZZ.

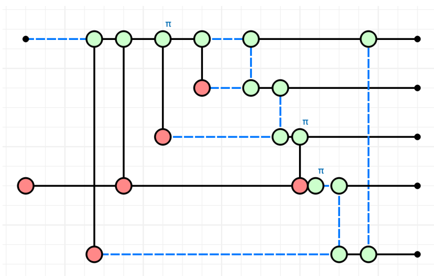
Third Pauli String A Pauli Z is attached to the 3rd qubit, and propogated through. Snapshots of the steps are below.



Step 1



Step 2



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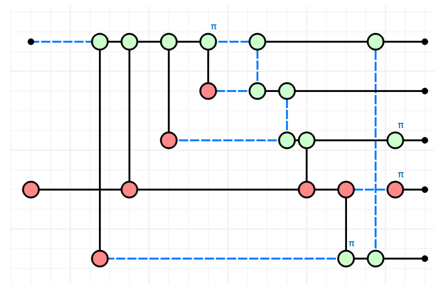

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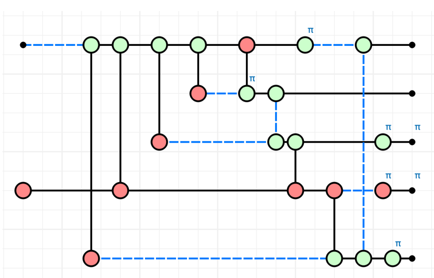

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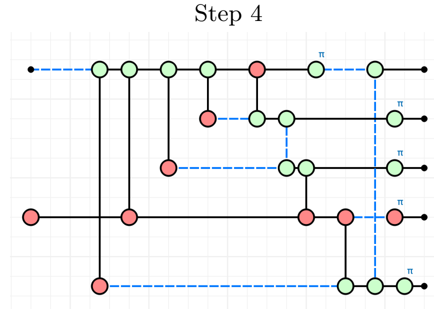
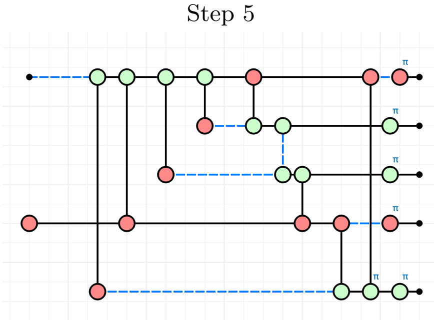
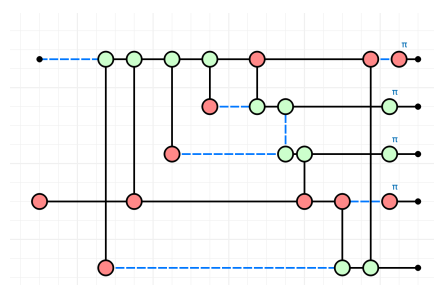


Step 6



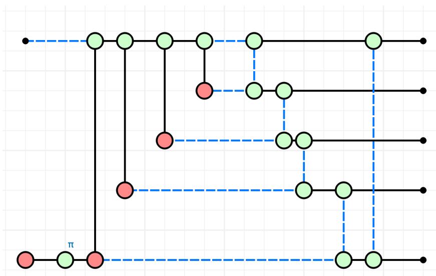
Step 7



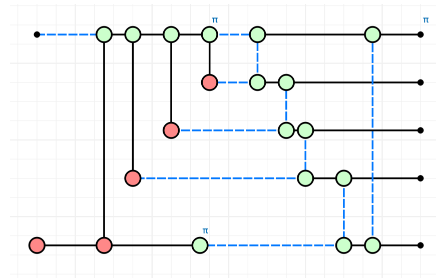
Step 8

For this string, there are 2 Pauli X gates on qubits 1 and 4, and 2 Pauli Z gates on qubits 2 and 3. The 5th qubit has the identity wire, which means that the 3rd Pauli String is XZZXI.

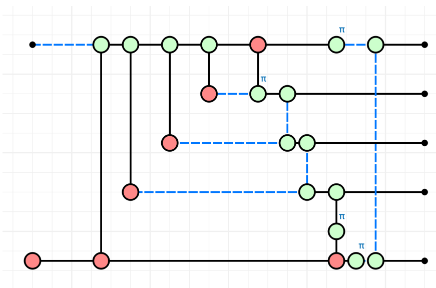
Fourth Pauli String Finally, the last stabilizer.



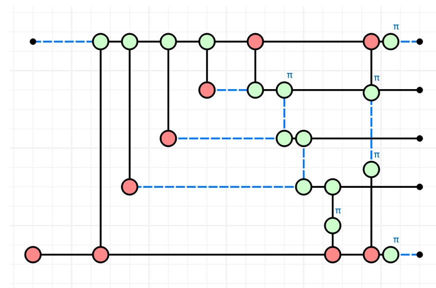
Step 1



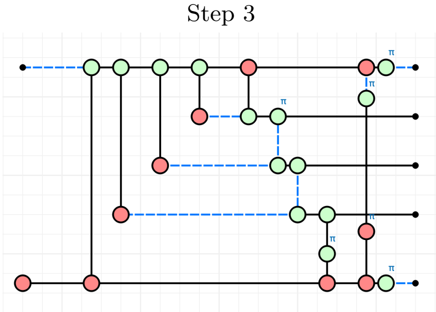
Step 2



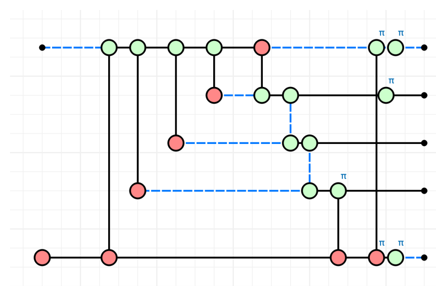
Step 3



Step 4



Step 5



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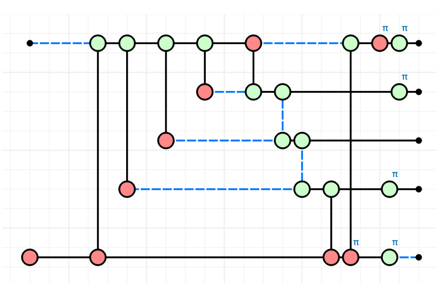

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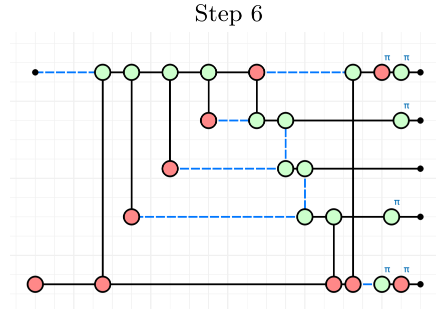


Step 8

Here, we have iY and -iY on qubits 1 and 5. There are 2 Pauli Z gates on qubits 2 and 4, and the 3rd qubit is identity. Therefore, the final stabilizer Pauli String is YZIZY.

Independence Together, all 4 Pauli strings are enumerated as such: YYZIZ, XIXZZ, XZZXI, and YZIZY. It needs to be shown that these generators are independent, meaning that they all commute with each other, and cannot be written as a combination of each other. Here, [A, B] = 0 if A and B commute.

$$[YYZIZ, XIXZZ] = 0 \quad \text{Y/X - ac. Y/I - c. Z/X - ac. I/Z - c. Z/Z - c.}$$
$$[YYZIZ, XZZXI] = 0 \quad \text{Y/X - ac. Y/Z - ac. Z/Z - c. I/X - c. Z/I - c.}$$
$$[YYZIZ, YZIZY] = 0 \quad \text{Y/Y - c. Y/Z - ac. Z/I - c. I/Z - c. Z/Y - ac.}$$
$$[XIXZZ, XZZXI] = 0 \quad \text{X/X - c. I/Z - c. X/Z - ac. Z/X - ac. Z/I - c.}$$
$$[XIXZZ, YZIZY] = 0 \quad \text{X/Y - ac. I/Z - c. X/I - c. Z/Z - c. Z/Y - ac.}$$
$$[XZZXI, YZIZY] = 0 \quad \text{X/Y - ac. Z/Z - c. Z/I - c. X/Z - ac. I/Y - c.}$$

Next we'll enumerate the rest of the stabilizers in the group, for a total of 16.

$$X \cdot Y = iZ \quad Y \cdot X = -iZ \quad Y \cdot Z = iX \quad Z \cdot Y = -iX \quad Z \cdot X = iY \quad X \cdot Z = -iY$$

$$YYZIZ * XIXZZ = ZYYZI$$
$$YYZIZ * XZZXI = ZXIXZ$$
$$YYZIZ * YZIZY = IXZZX$$
$$XIXZZ * XZZXI = IZYYZ$$
$$XIXZZ * YZIZY = ZZXIX$$
$$XZZXI * YZIZY = ZIZYY$$

$$YYZIZ * XIXZZ * XZZXI = YXXYI$$

$$YYZIZ * XIXZZ * YZIZY = XXYIY$$

$$YYZIZ * XZZXI * YZIZY = XYIYX$$

$$XIXZZ * XZZXI * YZIZY = YIYXX$$

$$YYZIZ * XIXZZ * XZZXI * YZIZY = IYXXY$$

Lastly, the trivial group IIIII

Problem 1.4

The code distance as defined in lecture is the minimum weight Pauli string that's greater than 0 that commutes with all strings inside the Stabilizer Group, but it itself does not exist within the Stabilizer Group.

$$\text{Smallest Weight of Pauli } \vec{P} \notin S . \forall \vec{Q} \in S, \vec{P}\vec{Q} = \vec{Q}\vec{P}$$

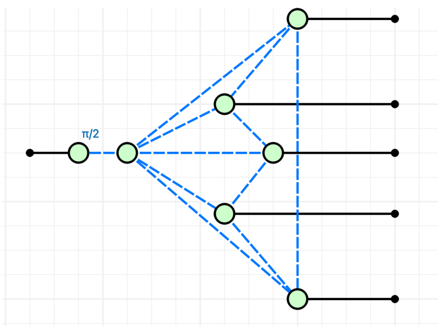
For the code distance to be 1, there must be a column where all elements commute with one another. Clearly, this is not possible, as every column contains X, Y, and Z, which inherently do not all commute.

For the code distance to be 2, a Pauli string must either commute or anti-commute with all stabilizers within both the elevant columns. Since no column contains only commuting elements, it is impossible for two columns to both commute entirely. The only remaining possibility is for both columns to anti-commute. However, in the given set of Pauli strings of length 5, each column (when considering the strings collectively) contains at least one I, which commutes with all other operators. This ensures that for every position in the strings, there is always at least one Pauli string in the set that does not commute with the others at that specific position. This makes it impossible to construct a Pauli string that anti-commutes in both positions.

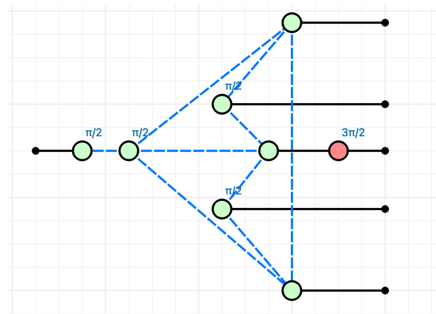
An example of a Pauli string that commutes with all stabilizers but is not part of the stabilizer group is IIYZY. Therefore, the code distance of the stabilizer code is 3.

Problem 1.5

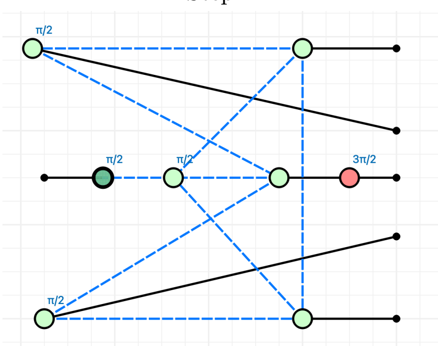
The idea is that E, given an S gate at the start, is a graph state. This means that local complementarity can be used. Step 1 is the initial diagram, and in Step 2 local complementarity is used on the middle node to the right. This means that local Cliffords are needed to adjust the diagram, adding ZSpider(pi/2) to it's neighbors and XSpider(-pi/2) to itself. Step 3 reorganizes the diagram visually, to prepare it to delete the middle spider. Step 4 deletes the middle spider, and since the deleted spider had a phase of pi/2, it subtracts pi/2 from it's neighbors. Lastly, the diagram is reorganized to show that only single qubit gates are needed for the physical qubits.



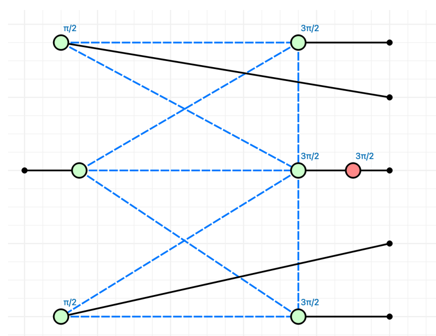
Step 1



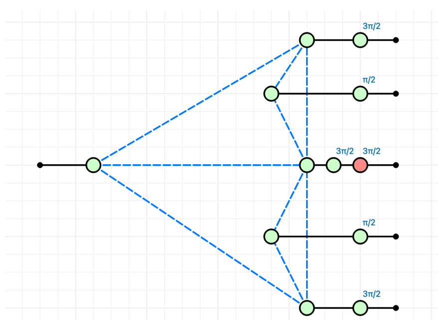
Step 2



Step 3



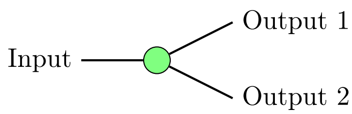
Step 4



Step 5

Problem 2.1

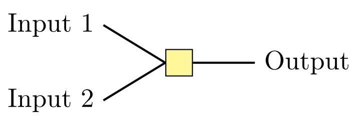
We need to prove that K_0 and K_1 are projectors, and that K forms a Von Neumann Measurement. The circuit diagram is first broken down into each individual component. The outer two Z-spiders each take one qubit and duplicates them.



Each Z-spider is therefore $|00\rangle\langle 0| + |11\rangle\langle 1|$, so the overall diagram is $(|00\rangle\langle 0| + |11\rangle\langle 1|) \otimes (|00\rangle\langle 0| + |11\rangle\langle 1|)$. This simplifies to the following:

$$|0000\rangle\langle 00| + |0011\rangle\langle 01| + |1100\rangle\langle 10| + |1111\rangle\langle 11|$$

The outer two qubits bypass any interaction and proceed directly to the output. The HBox is combined with identity operations on both sides, represented as $I \otimes \text{HBox} \otimes I$, before interacting with the outputs of the Z-spiders. Consequently, the HBox operates exclusively on the inner two qubits. It receives the states $|00\rangle$, $|10\rangle$, $|01\rangle$, and $|11\rangle$. Let us now break down the HBox with a phase of -1.



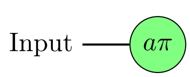
Based off the definition 9.8, 9.9 in Picturing Quantum Software, the matrix and ket-bra definitions for the HBox are below.

$$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

$$\frac{1}{\sqrt{2}}\sum a^{i_1 \dots i_m j_1 \dots j_n} |j_1 \dots j_n\rangle\langle i_1 \dots i_m|$$

$$\text{HBox} = \frac{1}{\sqrt{2}}(|0\rangle\langle 00| + |0\rangle\langle 01| + |0\rangle\langle 10| + |0\rangle\langle 11| + |1\rangle\langle 00| + |1\rangle\langle 01| + |0\rangle\langle 10| - |1\rangle\langle 11|)$$

We can see how this linear maps work. Inputting states $|00\rangle$, $|10\rangle$, $|01\rangle$ will result in $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$. Inputting the states $|11\rangle$ results in the Hbox mapping it to $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$. Now, we need to analyze the phase gate, which introduces a global phase only when certain inputs are detected.



$$\text{Z-Spider}: [1, 1] \text{ when } \alpha = 0$$
$$\text{Z-Spider}: [1, -1] \text{ when } \alpha = 1$$

The matrix definitions for the + and - states are as follows:

$$|+\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad |-\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

When $\alpha = 0$ or $\alpha = 1$, the results for the input states are given by:

$$\text{Inputting } |+\rangle = \begin{cases} \sqrt{2}, & \text{if } \alpha = 0, \\ 0, & \text{if } \alpha = 1, \end{cases}$$

and through simple matrix multiplication.

$$\text{Inputting } |-\rangle = \begin{cases} 0, & \text{if } \alpha = 0, \\ \sqrt{2}, & \text{if } \alpha = 1, \end{cases}$$

A constant of 0 is ignored, and for all other cases the global phase of $\frac{1}{\sqrt{2}}$ outside the circuit diagram cancels out the $\sqrt{2}$ to equal 1. Putting all the pieces together:

$$K_0 = |00\rangle\langle 00| + |10\rangle\langle 10| + |01\rangle\langle 01|$$

$$K_1 = |11\rangle\langle 11|$$

Now, by the definition of a projector, we need to show that both K_0 and K_1 are self-adjoint and idempotent.

$$K_0 = K_0^\dagger \text{ and } K_0 = K_0^2$$
$$K_1 = K_1^\dagger \text{ and } K_1 = K_1^2$$

By reversing the kets and bras, we find the complex conjugate. Clearly, they are equal. Taking the squares of these linear maps is also trivially equal since it maps the inputs back to themselves. Doing the same mapping twice won't change the outcome, which is why both K_0 and K_1 are equal to their squares.

$$K_0^\dagger = |00\rangle\langle 00| + |10\rangle\langle 10| + |01\rangle\langle 01| = K_0 = K_0^2$$
$$K_1^\dagger = |11\rangle\langle 11| = K_1 = K_1^2$$

Finally, we need to show that K a Von Neumann Measurement, which means that $K_0 + K_1 = K = I$. The final linear map always maps the inputs back to itself, which is trivially the identity map.

$$K_0 + K_1 = |00\rangle\langle 00| + |10\rangle\langle 10| + |01\rangle\langle 01| + |11\rangle\langle 11| = I$$

Problem 2.2

The image that this projector projects upon is a subspace that's spanned by basis vectors $|00\rangle$, $|10\rangle$, $|01\rangle$, $|11\rangle$. In other words, the subspace is $\mathbb{C}^2 \otimes \mathbb{C}^2$. Any state $|\phi\rangle$ can be written as follows:

$$|\phi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

Problem 2.3

To find the probability of each outcome under the state $|{+}{+}\rangle$, we need to use the borne rule. This means sandwiching K_0 or K_1 in between the states and effects of $++$. The inner product $\langle ab|cd\rangle$ equals 1 if and only if the first qubit states are the same ($|a\rangle = |c\rangle$) and the second qubit states are the same ($|b\rangle = |d\rangle$). Otherwise, it equals 0.

$$\text{prob} = \langle{+}{+}| K_0 |{+}{+}\rangle$$
$$= \frac{1}{2}(\langle 00| + \langle 01| + \langle 10| + \langle 11|) * (|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10|) + \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$
$$= \frac{1}{4}(\langle 00|00\rangle + \langle 01|01\rangle + \langle 10|10\rangle) = \frac{3}{4}$$

$$\text{prob} = \langle ++| \, K_1 \, |++\rangle$$
$$= \frac{1}{2}(\langle 00| + \langle 01| + \langle 10| + \langle 11|) * (|11\rangle \langle 11|) + \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$
$$= \frac{1}{4}\langle 11|11\rangle = \frac{1}{4}$$

Ultimately, we see that the probability for K_0 under state $|++\rangle$ is $\frac{3}{4}$ and the probability for K_1 is $\frac{1}{4}$.

Problem 2.4

The triangle operator is defined with a X-spider (pi phase), and an HBox that always has a $|+\rangle$ state as one of it's inputs. The X-spider has a rotation about the X-axis in the Bloc Sphere, which means that it turns the state $|0\rangle \to |1\rangle$ and $|1\rangle \to |0\rangle$. $|+\rangle$ remains unchanged and $|-\rangle \to |1\rangle - |0\rangle$ (a relative phase change of -1). Let's enumerate what happens for the triangle pointing right, ignoring global phases.

$$|0\rangle \to_{XSpider} |1\rangle$$
$$= |1\rangle \otimes |+\rangle = |10\rangle + |11\rangle \to_{HBox} |+\rangle + |-\rangle = |0\rangle + |1\rangle + |0\rangle - |1\rangle = 2 \cdot |0\rangle = |0\rangle \tag{1}$$

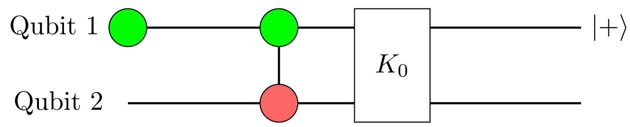
$$|1\rangle \to_{XSpider} |0\rangle$$

$$= |0\rangle \otimes |+\rangle = |00\rangle + |01\rangle . \to_{HBox} |+\rangle + |+\rangle = |0\rangle + |1\rangle + |0\rangle + |1\rangle = 2 \cdot |+\rangle = |+\rangle \tag{2}$$

$$|-\rangle \to_{XSpider} |-\rangle$$
$$= |-\rangle \otimes |+\rangle = |00\rangle + |10\rangle - |01\rangle - |11\rangle . \to_{HBox}\to |+\rangle + |+\rangle - |+\rangle - |-\rangle$$
$$= |0\rangle + |1\rangle - |0\rangle + |1\rangle = 2 \cdot |1\rangle = |1\rangle \tag{3}$$

$$|+\rangle \to_{XSpider} |+\rangle$$
$$= |+\rangle \otimes |+\rangle = |00\rangle + |01\rangle + |10\rangle + |11\rangle . \to_{HBox}\to |+\rangle + |+\rangle + |+\rangle + |-\rangle$$
$$= |0\rangle + |1\rangle + |0\rangle + |1\rangle + |0\rangle + |1\rangle + |0\rangle - |1\rangle = 2 \cdot (|+\rangle + |0\rangle = |+\rangle + |0\rangle \tag{4}$$

To stimulate this circuit, we focus on inputs $|0\rangle$ and $|1\rangle$, which yields the states $|0\rangle$ and $|+\rangle$ respectively. We'll need to entangle 2 qubits, and to make it nondeterministic when we take the measurement, a $|+\rangle$ state is prepared as the auxillary qubit since it's in superposition. The circuit for the triangle pointing right is shown on the next page.



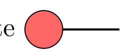
This diagram simulates the triangle pointing to the right, as the auxiliary qubit is prepared in the $|+\rangle$ state, placing it in a superposition. When the input qubit passes through the CNOT gate, there is a $\frac{1}{2}$ probability that it will be flipped by the target X-spider. The K_0 measurement is used to filter out the $|11\rangle$ state. Since the $|11\rangle$ state is ignored by post-selecting $|00\rangle$, this always maps 0 to 0 when the procedure is valid.

When the input qubit is 1, there is a $\frac{1}{2}$ probability of ending up in the $|10\rangle$ state and a $\frac{1}{2}$ probability of ending up in the $|01\rangle$ state. The K_0 measurement is unable to distinguish any meaningful information since $|10\rangle$ and $|01\rangle$ are both valid. Since qubit 2 is equally likely to be $|0\rangle$ or $|1\rangle$, it's effectively in a superposition, equivalent to the $|+\rangle$ state. Overall, this procedure fails if the $|11\rangle$ state is produced by the CNOT.

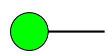
Idk how to do the flipped side lol.

Problem 2.5

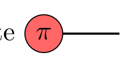
An X-Spider with no phase corresponds to the $|0\rangle$ state. As shown in Equation (1), this is equivalent to the $|0\rangle$ state. Thus, it results in an X-Spider with no inputs, one output, and no phase.

$|0\rangle$ State 

An X-Spider with a phase of π corresponds to the $|1\rangle$ state. After passing through the triangle, as shown in Equation (2), this becomes equivalent to the $|+\rangle$ state. Consequently, it results in a Z-Spider with no inputs, one output, and no phase.

$|+\rangle$ State 

A Z-Spider with a phase of π corresponds to the $|-\rangle$ state. As shown in Equation (3), this is equivalent to the $|1\rangle$ state after passing through the triangle. This is the same as an X-Spider with no inputs, one output, and a phase of π.

$|1\rangle$ State 

Problem 2.6

Let's understand what the Z-spiders in the diagram do, then we'll enumerate each combination of input to show that it's equal to the AND gate. For the Z-Spider with one input/output and a phase of pi, it's a rotation about the Z-axis. This means that the $|0\rangle$ state remains unchanged, and a global phase is introduced to $|1\rangle$. However, $|+\rangle \to |-\rangle$ and $|-\rangle \to |+\rangle$. For the Z-Spider with 2 inputs and 1 output, the ket/bra formula is written below.

$$S_Z^{(2,1)}(\alpha) = \sum_{x\in\{0,1\}} |x\rangle\langle x| \otimes \langle x| \cdot e^{i\alpha x}$$

$$= S_Z^{(2,1)}(\pi) = |0\rangle\langle 0| \otimes \langle 0| - |1\rangle\langle 1| \otimes \langle 1|$$

$$= S_Z^{(2,1)}(\pi) = |0\rangle\langle 00| - |1\rangle\langle 11|$$

All four possible combinations are enumerated below. ZSpider(2) is going through the 2 legged Z-Spider with pi phase, and ZSpider(1) is going through the 1 legged Z-Spider with pi phase. We can clearly see that the diagram is logically equivalent to the AND gate, QED.

$$|00\rangle \rightarrow_{Triangle} = |00\rangle \rightarrow_{ZSpider(2)} = |0\rangle \rightarrow_{triangle} = |0\rangle \rightarrow_{ZSpider(1)} = |0\rangle$$

$$|01\rangle \rightarrow_{Triangle} = |0\rangle \otimes |+\rangle = |00\rangle + |01\rangle \rightarrow_{ZSpider(2)} = |0\rangle \rightarrow_{triangle} = |0\rangle \rightarrow_{ZSpider(1)} = |0\rangle$$

$$|10\rangle \rightarrow_{Triangle} = |+\rangle \otimes |0\rangle = |00\rangle + |10\rangle \rightarrow_{ZSpider(2)} = |0\rangle \rightarrow_{triangle} = |0\rangle \rightarrow_{ZSpider(1)} = |0\rangle$$

$$|11\rangle \rightarrow_{Triangle} = |+\rangle \otimes |+\rangle = |00\rangle + |01\rangle + |10\rangle + |11\rangle \rightarrow_{ZSpider(2)} = |0\rangle - |1\rangle = |-\rangle \rightarrow_{triangle} = |1\rangle \rightarrow_{ZSpider(1)} = |1\rangle$$

Problem 2.7

The idea is that the CCZ is non-Pauli and non-Clifford, and being able to simulate it with Pauli Measurements, Clifford Gates, and K-Measurements means that combination of those 3 are computationally universal. Probably some entangled state of 3 qubits in $|+\rangle$ states with a series of CNOTS, but idk.